

JAVA INTERFACE PAYROLL PROGRAM

Object-Oriented Programming Practical Record

WEEK 8

Question - Interfaces - Display monthly salary details for regular and contract employees. A regular employee has Basic Pay 25000, HRA 15000 and TA 5000; a contract employee has Basic Pay 12000 and TA 3000 with no HRA.

Concept: Interfaces and accessing implementations through interface references.

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1. Steps for Execution

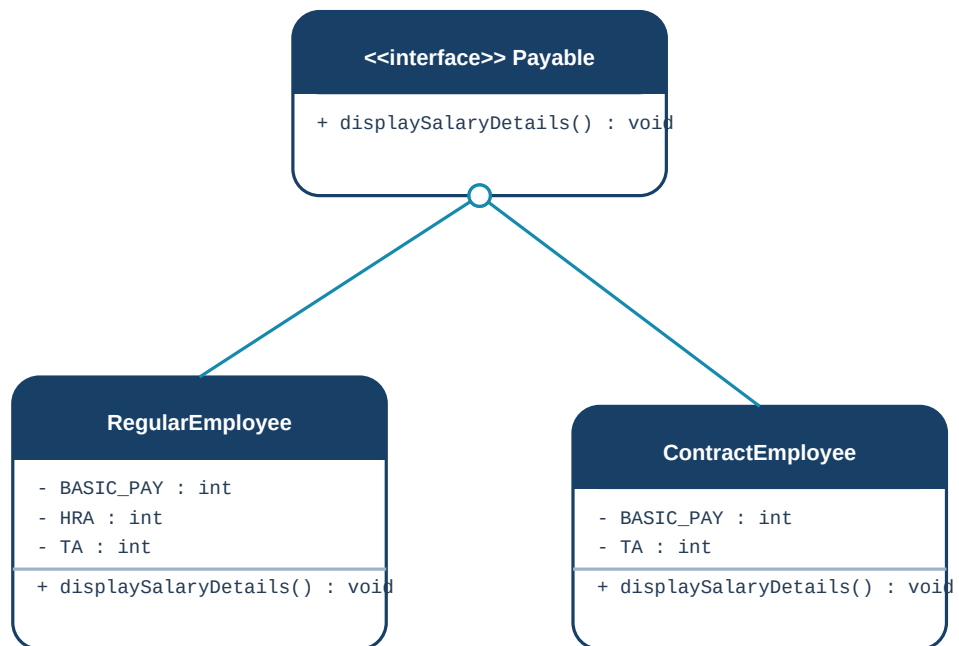
1. Open a Java IDE or command prompt with the Java Development Kit installed.
2. Create a source file named `Week8EmployeePayrollInterface.java`.
3. Copy the program into the file and save it.
4. Compile it using `javac Week8EmployeePayrollInterface.java`.
5. Run it using `java Week8EmployeePayrollInterface`.
6. Provide the required input when prompted and observe the result.

2. Algorithm

1. Define a `Payable` interface with `displaySalaryDetails()`.
2. Implement it in `RegularEmployee` and `ContractEmployee` classes.
3. Read the employee type from the user.
4. Create the appropriate object through a `Payable` reference and display its salary.

3. UML Class Diagram

The diagram shows the classes and their principal attributes and methods used in this program.



4. Java Source Code

Filename: Week8EmployeePayrollInterface.java

```
import java.util.Scanner;

interface Payable {
    void displaySalaryDetails();
}

class RegularEmployee implements Payable {
    private static final int BASIC_PAY = 25000;
    private static final int HRA = 15000;
    private static final int TA = 5000;
    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d\nHRA: %d\nTA: %d\nTotal Amount: %d\n",
            BASIC_PAY, HRA, TA, BASIC_PAY + HRA + TA);
    }
}

class ContractEmployee implements Payable {
    private static final int BASIC_PAY = 12000;
    private static final int TA = 3000;
    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d\nHRA: 0\nTA: %d\nTotal Amount: %d\n",
            BASIC_PAY, TA, BASIC_PAY + TA);
    }
}

public class Week8EmployeePayrollInterface {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter employee ID (R for Regular, C for Contract): ");
        String type = scanner.nextLine().trim().toUpperCase();
        Payable employee = type.startsWith("C") ? new ContractEmployee() : new RegularEmployee();
        employee.displaySalaryDetails();
        scanner.close();
    }
}
```

5. Representative Sample Output

```
Enter employee ID (R for Regular, C for Contract): R
Basic Pay: 25000
HRA: 15000
TA: 5000
Total Amount: 45000
```

Result

Thus, the Java program was written and executed successfully.